

Name: kamanisairam

Reg no:192312195

Ex.No 11: 5G Network Slicing Environment Using MATLAB

Objective 1:

To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB..

Program (Matlab Code)

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')

grid on
hold on
```



```
ylabel(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
```

```
status = soil_moisture >= threshold;
```

```
subplot(2,1,2)
```

```
stem(time, status, 'filled', 'LineWidth', 2)
```

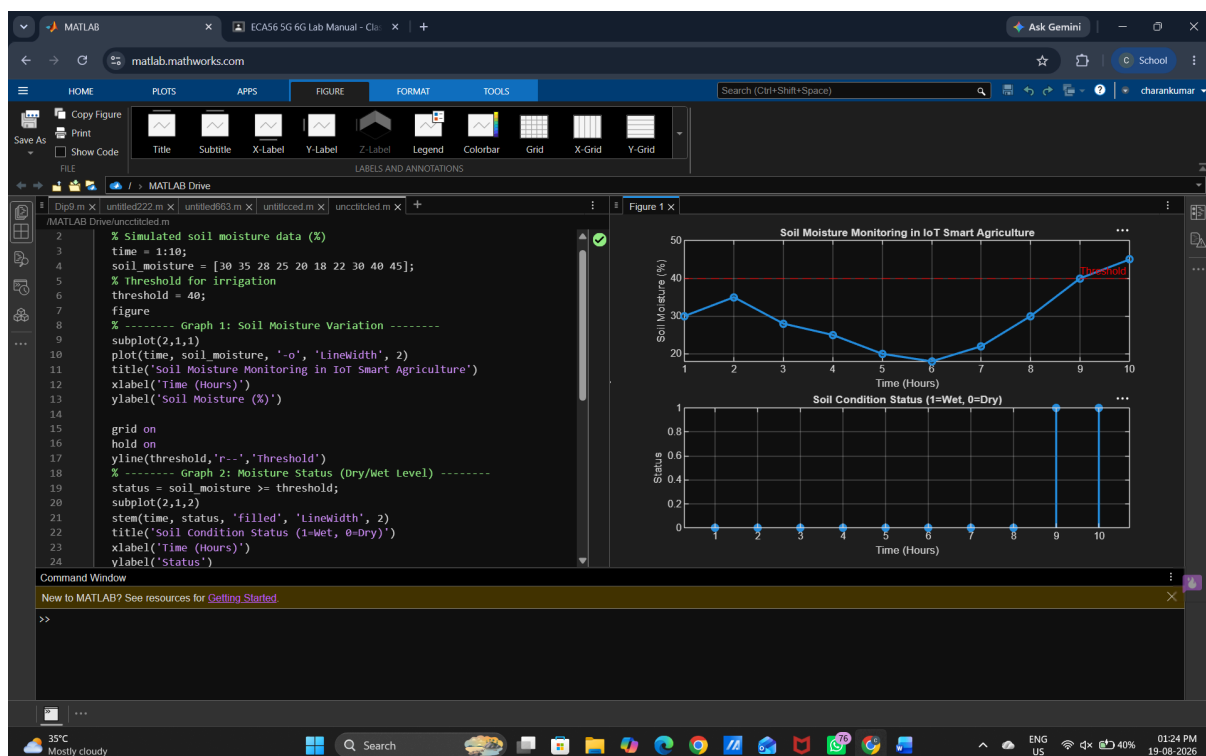
```
title('Soil Condition Status (1=Wet, 0=Dry)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Status')
```

```
grid on
```

Output:



Objective 2:

To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions..

Matlab Code

```
clc; clear; close all;
% Simulated soil moisture (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% ----- Graph 1: Soil Moisture vs Pump Status -----
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
```



```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```
ylabel(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Pump Status (ON/OFF) -----
```

```
subplot(2,1,2)
```

```
stem(time, pump_status, 'filled','LineWidth',2)
```

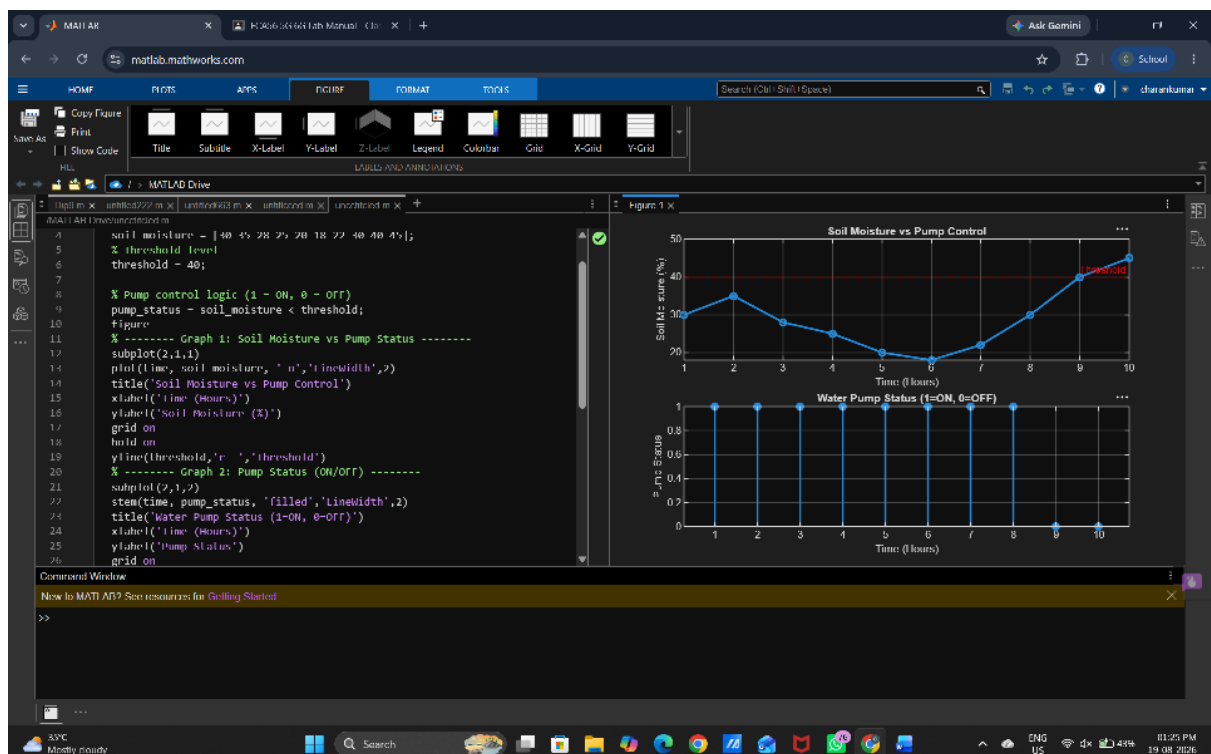
```
title('Water Pump Status (1=ON, 0=OFF)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Pump Status')
```

```
grid on
```

Output



Objective 3:

To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level..

Matlab Code

```
clc; clear; close all;  
% Simulated soil moisture data (%)  
time = 1:10;  
soil_moisture = [30 35 28 25 20 18 22 30 40 45];  
% Threshold level  
threshold = 40;  
% Irrigation efficiency calculation (%)  
efficiency = (soil_moisture ./ threshold) * 100;  
efficiency(efficiency > 100) = 100; % limit to 100%  
figure  
% ----- Graph 1: Soil Moisture vs Efficiency -----  
subplot(2,1,1)
```



```
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% ----- Graph 2: Irrigation Efficiency -----
subplot(2,1,2)
plot(time, efficiency, '-s','LineWidth',2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```



Output

